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Subject: Mathematics

Class : IX

Set : II

Section : C

Roll no. : 9339

Section - A

Solⁿ 1] (c)

Solⁿ 2] (d)

Solⁿ 3] (a)

Solⁿ 4] (d)

Solⁿ 5] (a)

Solⁿ 6] (c)

Solⁿ 7] (c)

Solⁿ 8] (b)

Solⁿ 9] (c)

Solⁿ 14] These coordinates are parallel to y-axis.

Because, the line they have formed is parallel to y-axis and both the equation contain 2 as its x-axis coordinate which makes it a straight line who's parallel to y-axis.

Section-c

Solⁿ 15] We have,

$$\frac{2}{5} \times \frac{10}{10} = \frac{20}{50} ; \frac{3}{5} \times \frac{10}{10} = \frac{30}{50}$$

$\therefore$ Three rational numbers = $\frac{21}{50}, \frac{22}{50}, \frac{23}{50}$

Solⁿ 17] $54a^3 - 250b^3$

$$= 2 [27a^3 - 125b^3]$$

$$= 2 [(3a)^3 - (5b)^3]$$

$$= 2 [(3a-5b)((3a)^2 + (3a)(5b) + (5b)^2)]$$

$$= 2 [(3a-5b)(9a^2 + 15ab + 25b^2)]$$

$$\{x^3 - y^3 = (x-y)(x^2 + xy + y^2)\}$$

Solⁿ 18] By Factor Theorem,

$$p(y) = 2y^3 + y^2 + 4y - 15$$

$$2y - 3 = 0$$

$$y = \frac{3}{2}$$

Putting $y = \frac{3}{2}$

$$p\left(\frac{3}{2}\right) = 2\left(\frac{3}{2}\right)^3 + \left(\frac{3}{2}\right)^2 + 4\left(\frac{3}{2}\right) - 15$$

$$= 2\left(\frac{27}{8}\right) + \frac{9}{4} + 6 - 15$$

$$= \frac{27}{4} + \frac{9}{4} - 9$$

$$= \frac{27+9-36}{4}$$

$$= \frac{36-36}{4}$$

$$= \frac{0}{4} = 0$$

$\therefore p(y)$ is a multiple of $(2y-3)$.

Solⁿ]

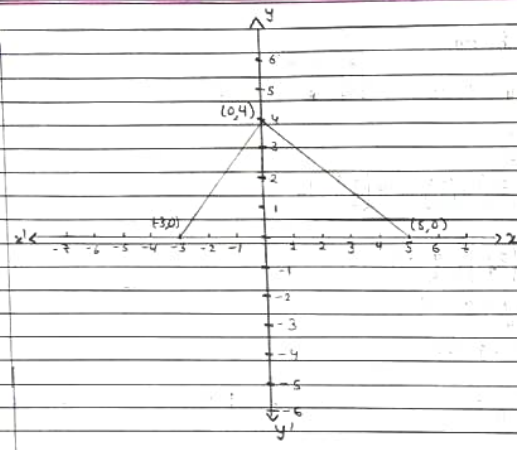


Figure: Triangle

b: 8 unit
h: 4 unit
Area: $\frac{1}{2} \times b \times h$
 $= \frac{1}{2} \times 8 \times 4 = 16 \text{ sq. unit}$

Solⁿ]

Given: $\angle 1 = \angle 2$, ABCD is a rectangle
 $\angle 3 = \angle 4$

To Prove: ABCD is a square

Proof: We have,

$\angle 1 = \angle 4$ } Alternate $\angle$ s
 $\angle 2 = \angle 3$ } Angles

And,

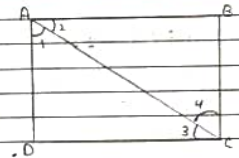
$\angle 1 = \angle 2$ } Given - (i)
 $\angle 3 = \angle 4$ }

From (i) and (ii)

$\angle 1 = \angle 2 = \angle 3 = \angle 4$

Since, a quadrilateral with all sides equal and $\angle$ s angles = 90° .
 $\therefore$ ABCD is a square

Hence Proved



Sol ⁿ 23	$a = 18 \text{ cm}$ $b = 10 \text{ cm}$ $c = ?$ Perimeter = 42	$s = \frac{\text{perimeter}}{2}$ $= \frac{42}{2} = 21 \text{ cm}$
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$$\therefore P = a + b + c$$

$$42 = 18 + 10 + c$$

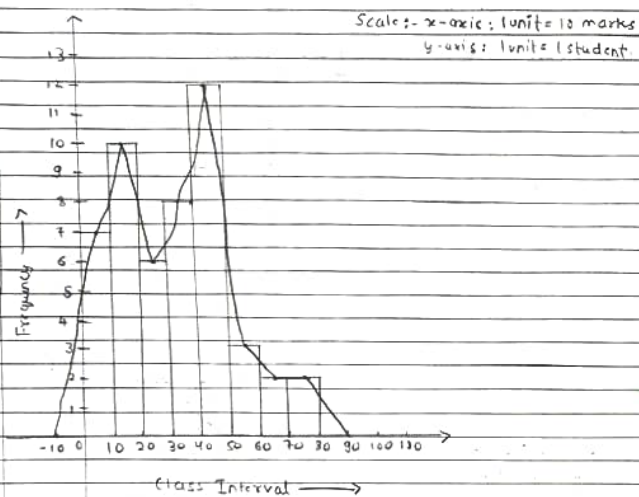
$$c = 42 - 28 \therefore c = 14 \text{ cm}$$

Now,

$$\begin{aligned} \text{Area of } \Delta &= \sqrt{s(s-a)(s-b)(s-c)} \\ &= \sqrt{21(21-18)(21-10)(21-14)} \\ &= \sqrt{21(3)(11)(7)} \\ &= 21\sqrt{11} \text{ cm}^2 \end{aligned}$$

Sol ⁿ 24	$h = 16 \text{ cm}$ $r = 12 \text{ cm}$ $l = \sqrt{r^2 + h^2}$ $l = \sqrt{12^2 + 16^2}$ $l = \sqrt{400}$ $l = 20 \text{ cm}$	$\therefore \text{T.S.A} = \pi r(r+l)$ $= \frac{22}{7} \times 12 (12+20)$ $= \frac{22}{7} \times 12 \times 32$ $= 1206.857 \text{ cm}^2$
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Sol ⁿ 27	Section - D	
	Marks obtained (C.I.)	No. of students (f)
	0-10	7
	10-20	10
	20-30	6
	30-40	8
	40-50	12
	50-60	3
	60-70	2
	70-80	2



Solⁿ: Given: $BC = AD$
 $\angle 3 = \angle 4 = 90^\circ$

To Prove: CD bisects AB

Proof: In $\triangle OBC$ and $\triangle OAD$

$\angle 1 = \angle 2$ (V.O.A)

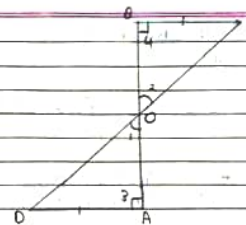
$\angle 3 = \angle 4$ (Each 90°)

$AD = BC$ (Given)

$\therefore \triangle OBC \cong \triangle OAD$ (By AAS)

$\therefore OD = OC$ (By C.P.T)

Hence, CD bisects AB



Solⁿ: Given: $AB = CD$

To Prove: $\angle AOB = \angle COD$

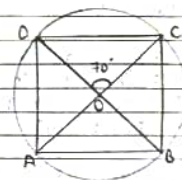
Proof: In $\triangle AOB$ and $\triangle COD$

$AB = CD$ (Given)

$OA = OB$ } radii of a circle

$OC = OD$ }

$\therefore \triangle AOB \cong \triangle COD$ (By S.S.S)



$$\therefore \angle AOB = \angle COD \text{ (By CPCT)}$$

Hence Proved

Now,

$$\angle COD = \angle AOB = 70^\circ \text{ (Proved above)}$$

And,

$OA = OB$ (radii of a circle)

$$\Rightarrow \angle OAB = \angle OBA \text{ (Theorem 7.2)}$$

$$\therefore \angle AOB + \angle OAB + \angle OBA = 180^\circ \text{ (Angle sum property)}$$

$$70^\circ + \angle OAB + \angle OAB = 180^\circ [\because \angle OAB = \angle OBA]$$

$$2\angle OAB = 180^\circ - 70^\circ$$

$$\angle OAB = 55^\circ \Rightarrow \angle OAB = \angle OBA = 55^\circ$$

$$\therefore \angle OAB = \angle OBA = 55^\circ$$

$$\text{Sol } 30] 8x^3 + 27y^3 + 36x^2y + 54xy^2$$

$$(2x)^3 + (3y)^3 + 3(2x)^2(3y) + 3(2x)(3y)^2$$

$$(2x)^3 + (3y)^3 + 3(2x)(3y)[2x+3y]$$

$$(2x)^3 + (3y)^3 + 18xy[2x+3y]$$

$$(2x+3y)^3$$

$$(2x+3y)(2x+3y)(2x+3y)$$

$$\left. \begin{aligned} \because (a+b)^3 &= a^3 + b^3 + 3ab(a+b) \\ &= a^3 + b^3 + 3a^2b + 3ab^2 \end{aligned} \right\}$$